Type MS · Hardware devices: input, output and how they connect

Answer 1 of 4 · Question 5 part b.i · 3 marks

2023 JUN · Paper 01 · extract 1 of 8 · find part b.i in this mark scheme extract

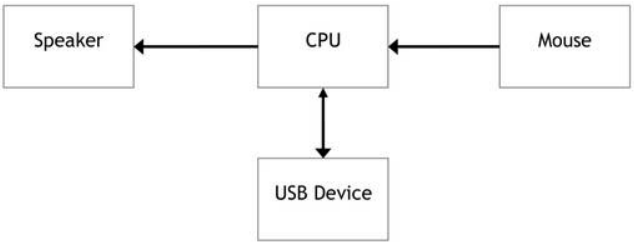
<table>
<tr>
<td>5(a)</td>
<td>Award one mark for each correct cell:
<table>
<tr><th>Req.</th><th>Example</th></tr>
<tr><td>Input</td><td>1. Enter the direction the bee will face (1)
2. Enter the number of steps that the bee will move (1)</td></tr>
<tr><td>Process</td><td>Calculate the path the bee will move along to its new position</td></tr>
<tr><td>Output</td><td>3. The bee appears in its new position (1)
4. The bee flies to its new position (1)
5. The bee faces its new direction (1)</td></tr>
</table>
</td>
<td>Allow direct instructions as input e.g. ‘Turn East’ and consequences for output e.g. ‘Bee turns to face East’ if clear.

The input/output do not have to be directly linked. E.g. Input Turn West, Output Has Moved 20 steps</td>
<td>2</td>
</tr>
</table>

Type MS · Hardware devices: input, output and how they connect

Answer 1 of 4 · Question 5 part b.i · 3 marks

2023 JUN · Paper 01 · extract 2 of 8 · find part b.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5b(i)	Award one mark for each: • Line between Mouse and CPU with arrow at CPU end only (1) • Line between CPU and Speaker with arrow at Speaker end only (1) • Line between CPU and USB Device with arrows on both ends (1) 	Allow two independent back/forth arrows between CPU and USB in place of a single bi-directional arrow. Ignore any additional lines.	3

Type MS · Hardware devices: input, output and how they connect

Answer 1 of 4 · Question 5 part b.i · 3 marks

2023 JUN · Paper 01 · extract 3 of 8 · find part b.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(b)(ii)	**The only correct answer is B** ***A** is not correct because a **firewall** is an application that is loaded after the device is booted* ***C** is not correct because **the operating system** is loaded after the device is booted* ***D** is not correct because **the user interface** is loaded after the device is booted*		1

Question Number	Answer	Additional Guidance	Mark
5(b)(iii)	Award up to **two** marks for a linked explanation, such as: • Cache makes up for the difference in speed of two devices (1) because it is very high-speed memory (1) • Cache is high-speed memory (1) which acts as a buffer between a faster and a slower device (1) • It speeds up processing (1) because active/commonly used instructions and data are stored nearer to the CPU / the CPU doesn't have to wait while instructions/data are fetched from RAM (1)		2

Type MS · Hardware devices: input, output and how they connect

Answer 1 of 4 · Question 5 part b.i · 3 marks

2023 JUN · Paper 01 · extract 4 of 8 · find part b.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(c)	Award up to **two** marks for a linked description, such as: • Uses a sensor (1) to check the water level in the machine (1) • Uses a thermometer (1) to check the water temperature (1) • Uses a timer (1) to control the time required for the wash cycle/rinse/drain/spin (1) • Uses a motor (1) to rotate the drum (1)	The first mark is for identification and the second is for the linked description. It is possible to gain one mark for a description without identifying the sensor/actuator.	2

Type MS · Hardware devices: input, output and how they connect

Answer 1 of 4 · Question 5 part b.i · 3 marks

2023 JUN · Paper 01 · extract 5 of 8 · find part b.i in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>5(d)(i)</td><td>Award one mark for each two correct rows.

<table>
<tr><th>A</th><th>B</th><th>C</th><th>A AND C</th><th>(A AND C) OR B</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr>
</table>
</td><td></td><td>3</td></tr>
</table>

Type MS · Hardware devices: input, output and how they connect

Answer 1 of 4 · Question 5 part b.i · 3 marks

2023 JUN · Paper 01 · extract 6 of 8 · find part b.i in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>5(d)(ii)</td><td>Award one mark for each of:

• M OR P (1)
• NOT C (1)
• <expr1> AND <expr2> (1)

Example statements:

(M OR P) AND (NOT C)
(M OR P) AND NOT C
(M AND NOT C) OR (P AND NOT C)
M AND NOT C OR P AND NOT C

For information only:

<table>
<tr><th>M</th><th>P</th><th>C</th><th>M OR P</th><th>NOT C</th><th>(M OR P) AND (NOT C)</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr>
</table>
</td><td>Third bullet awarded even if first two are not

Order of precedence is NOT, AND, OR

Check brackets carefully, if provided</td><td>3</td></tr>
</table>

Type MS · Hardware devices: input, output and how they connect

Answer 1 of 4 · Question 5 part b.i · 3 marks

2023 JUN · Paper 01 · extract 7 of 8 · find part b.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(e)	Award **one** mark for any of the following up to a maximum of **two** marks: 1. The operating system uses a scheduler to control processes (1) 2. The operating system holds processes in a queue (1) 3. Some processes may be given higher priorities than others (1) 4. Each process gains accesses to the CPU for a short time / time slice to execute (1) 5. Processes are swapped to/from (the queue / CPU)	Do not award responses about memory management, as the question is about sharing the CPU only	2

Type MS · Hardware devices: input, output and how they connect

Answer 1 of 4 · Question 5 part b.i · 3 marks

2023 JUN · Paper 01 · extract 8 of 8 · find part b.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Hardware devices: input, output and how they connect

Answer 2 of 4 · Question 3 part a · 2 marks

2021 JUN · Paper 01 · extract 1 of 2 · find part a in this mark scheme extract

Number		Guidance	
3(a)	Award **two** marks for an explanation that includes hardware **and** software. • The laptop is/contains the hardware (1) and the software allows the user to interact with it (1) • Hardware gives input and output devices (1) /memory, storage, cpu, processing ability • Software allows instructions to be passed to the hardware (1) so that it can carry out tasks (1) • Software is the set of instructions (1) that controls the hardware (1)		2

Question Number	Answer	Additional Guidance	Mark
3(b)(i)	Award **two** marks from: • Lighter/thinner (1) • Less heat generated (1) • Faster access speed / faster booting of operating system / faster data transfer/read/write speeds (1) • Silent operation (1) • Less power required / longer battery life (1) • More robust (due to no moving parts) (1)		2
3(b)(ii)	Award **two** marks for a linked explanation such as: • Flash memory is non-volatile (1) whereas RAM is volatile (1) • Flash memory persists even when power is lost (1) whereas RAM clears its data whenever power is lost (1) • Flash memory is used for storage (1) whereas RAM is used in performing operations on data taken from storage (1)		2

Type MS · Hardware devices: input, output and how they connect

Answer 2 of 4 · Question 3 part a · 2 marks

2021 JUN · Paper 01 · extract 2 of 2 · find part a in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
3(c)	Award **two** marks for a linked explanation such as: • Cache memory stores active/commonly used instructions / cache is a temporary storage (1) and is used to speed up processing. (1) stop data transfer bottlenecks between RAM and CPU (1)		2

Question Number	Answer	Additional Guidance	Mark
3(d)	Award **one** mark for two correct rows. Award **two** marks for four correct rows.		2

Device	Input	Output	Both
Headset			✓
Mouse	✓		
Printer		✓	
Touch screen			✓

Question Number	Answer	Additional Guidance	Mark
3(e)	Award **two** marks from: • Antivirus (1) • Anti-spyware (1) • (Software) firewall (1)	Do not award tradenames	2

Question Number	Answer	Additional Guidance	Mark

Type MS · Hardware devices: input, output and how they connect

Answer 3 of 4 · Question 3 part d · 2 marks

2021 JUN · Paper 01 · extract 1 of 2 · find part d in this mark scheme extract

Number		Guidance	
3(a)	Award **two** marks for an explanation that includes hardware **and** software. • The laptop is/contains the hardware (1) and the software allows the user to interact with it (1) • Hardware gives input and output devices (1) /memory, storage, cpu, processing ability • Software allows instructions to be passed to the hardware (1) so that it can carry out tasks (1) • Software is the set of instructions (1) that controls the hardware (1)		2

Question Number	Answer	Additional Guidance	Mark
3(b)(i)	Award **two** marks from: • Lighter/thinner (1) • Less heat generated (1) • Faster access speed / faster booting of operating system / faster data transfer/read/write speeds (1) • Silent operation (1) • Less power required / longer battery life (1) • More robust (due to no moving parts) (1)		2
3(b)(ii)	Award **two** marks for a linked explanation such as: • Flash memory is non-volatile (1) whereas RAM is volatile (1) • Flash memory persists even when power is lost (1) whereas RAM clears its data whenever power is lost (1) • Flash memory is used for storage (1) whereas RAM is used in performing operations on data taken from storage (1)		2

Type MS · Hardware devices: input, output and how they connect

Answer 3 of 4 · Question 3 part d · 2 marks

2021 JUN · Paper 01 · extract 2 of 2 · find part d in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
3(c)	Award **two** marks for a linked explanation such as: • Cache memory stores active/commonly used instructions / cache is a temporary storage (1) and is used to speed up processing. (1) stop data transfer bottlenecks between RAM and CPU (1)		2

Question Number	Answer	Additional Guidance	Mark
3(d)	Award **one** mark for two correct rows. Award **two** marks for four correct rows.		2

Device	Input	Output	Both
Headset			✓
Mouse	✓		
Printer		✓	
Touch screen			✓

Question Number	Answer	Additional Guidance	Mark
3(e)	Award **two** marks from: • Antivirus (1) • Anti-spyware (1) • (Software) firewall (1)	Do not award tradenames	2

Question Number	Answer	Additional Guidance	Mark

Type MS · Hardware devices: input, output and how they connect

Answer 4 of 4 · Question 2 part a · 2 marks

2025 JUN · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

2(a)	Award **one** mark for any of the following up to a maximum of **two** marks: • (Touch)screen (1) • Speaker (1) • Vibration (motor) (1) • Flashlight/LED light/torch (1) • NFC chip (1)	**Accept:** • display for screen **Do not accept:** • monitor • printer • headset • other peripheral devices	**2**
Question	Answer	Additional Guidance	Mark
2(b)(i)	**The only correct answer is D** *A is not correct because cache is located closer to the CPU* *B is not correct because random-access memory is a physical card on the motherboard* *C is not correct because read-only memory cannot be written to*		**1**
Question	Answer	Additional Guidance	Mark
2(b)(ii)	Award **one** mark for any of the following up to a maximum of **two** marks: • Often held/stored remotely/stored on the internet (1) • Held/managed by a third party (1) • Requires internet connectivity to access (1)		**2**

Type MS · Hardware devices: input, output and how they connect

Answer 4 of 4 · Question 2 part a · 2 marks

2025 JUN · Paper 01 · extract 2 of 3 · find part a in this mark scheme extract

Question	Answer	Additional Guidance	Mark
2(c)(i)	Award **one** mark for any of the following up to a maximum of **three** marks: • Clock speed/number of cycles per second (1) • Number of processor cores (1) • Size of cache (memory) (1) • Type of cache (1) • Width of data bus/word length (1) • Width of address bus (1) • Whether the CPU is overclocked or not (1) • Pipelining architecture (1) • Thermal throttling / cooling system (reducing clock speed) (1)	**Accept:** • units of clock speed e.g. GHz for MP1 • width of bus for either MP5 or MP6 but not both **Do not accept:** cache without clarification of size or type	3
Question	Answer	Additional Guidance	Mark
2(c)(ii)	Award up to **two** marks for a linked description such as: • The CPU places an address on to the address bus (1) to specify the location in main memory (1) • (The address bus) contains the location in RAM (1) to read from/write to (1) • The value in the memory address register (MAR) is placed on the address bus (1) to access the correct location in RAM (1) • The address bus carries the memory location (1) of the current/next instruction (1)	Explanation can be a combination of different marking points. E.g. MP1a and MP2b	2
Question	Answer	Additional Guidance	Mark
2(d)	Award **one** mark for any of the following up to a maximum of **two** marks: • Sub-atomic particles (1) • Qubit(s) (1) • Entanglement (1) behaviour of a pair of qubits can influence each other • Quantum parallelism (1) - allowing a quantum computer to evaluate multiple computational paths simultaneously	**Do not accept:** 'Qubits can exist in a state of 1 and 0 at the same time' as this is the same as 'superposition' given in the question	2
		Total for question 2	**12**

Type MS · Hardware devices: input, output and how they connect

Answer 4 of 4 · Question 2 part a · 2 marks

2025 JUN · Paper 01 · extract 3 of 3 · find part a in this mark scheme extract

Question	Answer	Additional Guidance	Mark